

作业纸

课程名称: _____

班级: _____

教学班级: _____

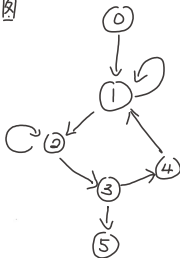
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学号: 1820021053

第 _____ 页

- 1 a) 块0 $d_1 - d_4$
 块1 $d_5 - d_8$
 块2 $d_9 - d_{12}$
 块3 d_{13}
 块4 $d_{14} - d_{21}$
 块5 $d_{23} - d_{30}$

程序流图



- b) $D(0) = \{0\}$
 $D(1) = \{0, 1\}$
 $D(2) = \{0, 1, 2\}$
 $D(3) = \{0, 1, 2, 3\}$
 $D(4) = \{0, 1, 2, 3, 4\}$
 $D(5) = \{0, 1, 2, 3, 5\}$

c) 回边

- $\langle 4, 1 \rangle$, $\langle 4, 1 \rangle_{loop} = \{4, 3, 2, 1\}$
 $\langle 1, 1 \rangle$, $\langle 1, 1 \rangle_{loop} = \{1\}$
 $\langle 2, 2 \rangle$, $\langle 2, 2 \rangle_{loop} = \{2\}$

联系方式: _____

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第 _____ 页

d)

块	Gen 集合	Kill 集合
0	$\{d_1, d_2, d_3, d_4\}$	$\{d_5, d_9\}$
1	$\{d_5, d_6, d_7\}$	$\{d_1\}$
2	$\{d_9, d_{10}, d_{11}\}$	$\{d_2\}$
3	$\emptyset$	$\emptyset$
4	$\{d_{14}, d_{15}, d_{16}, d_{17}, d_{18}, d_{19}, d_{20}, d_{21}\}$	$\emptyset$
5	$\{d_{23}, d_{24}, d_{25}, d_{26}, d_{27}, d_{28}, d_{29}\}$	$\emptyset$

e) 对于块 0, $IN_0 = \emptyset$

对于块 1, $IN_1 = \{d_1, d_2, d_3, d_4, d_5, d_6, d_7, d_{10}, d_{11}, d_{14}, d_{15}, d_{16}, d_{17}, d_{18}, d_{20}\}$

对于块 2, $IN_2 = \{d_1, d_3, d_4, d_5, d_6, d_7, d_9, d_{10}, d_{11}, d_{14}, d_{15}, d_{16}, d_{17}, d_{20}\}$

对于块 3, $IN_3 = \{d_3, d_4, d_5, d_6, d_7, d_9, d_{10}, d_{11}, d_{14}, d_{15}, d_{16}, d_{17}, d_{20}\}$

对于块 4, $IN_4 = IN_3$

对于块 5, $IN_5 = IN_4 = IN_3$

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第 _____

页

f) 块

DEF 集合

USE 集合

0 $\{i(5, 6, 13, 14, 16, 23, 25), j(9, 10, 13, 17, 20), v(8, 12)\}$

$\{n(1), n(2, 3), a(4)\}$

1 $\{i(13, 14, 16, 23, 25)\}$

$\{i(5), a(7), v(8)\}$

2 $\{j(13, 17, 20)\}$

$\{j(9), a(11), v(12)\}$

3 $\emptyset$

$\{i(13), j(13)\}$

4 $\emptyset$

$\{i(14, 16), j(17, 20), a(15, 18, 19, 21), x(21)\}$

5 $\{x(21)\}$

$\{i(23, 25), u(26, 29), a(24, 27, 28, 30)\}$

g) 块

OUT 集合

0 $\{i(15), j(9), a(7, 11, 15, 18, 19, 21, 24, 27, 28, 30), n(26, 29), v(8, 12), x(21)\}$

1 $\{i(5, 13, 14, 16, 23, 25), j(9), a(7, 11, 15, 18, 19, 21, 24, 27, 28, 30), n(26, 29), v(8, 12), x(21)\}$

2 $\{i(5, 13, 14, 16, 23, 25), j(9, 17, 20), a(7, 11, 15, 18, 19, 21, 24, 27, 28, 30), n(26, 29), v(8, 12), x(21)\}$

3 $\{i(5, 14, 16, 23, 25), j(9, 17, 20), a(7, 11, 15, 16, 19, 21, 24, 27, 28, 30), n(26, 29), v(8, 12), x(21)\}$

4 $\{i(5), j(9), a(7, 11, 15, 18, 19, 21, 24, 27, 28, 30), n(26, 29), v(8, 12), x(21)\}$

5 $\emptyset$

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第 _____ 页

h) $d_{5-(1)}ud = \{d_1, d_5\}$, $d_{6-(1)}ud = \{d_5\}$, $d_{7-(1)}ud = \emptyset$, $d_{7-(t_2)}ud = \{d_6\}$, $d_{8-(t_2)}ud = \{d_5\}$
 $d_{8-(v)}ud = \{d_4\}$, $d_{9-(1)}ud = \{d_2, d_4\}$, $d_{10-(1)}ud = \{d_4\}$, $d_{11-(t_4)}ud = \{d_{10}\}$, $d_{12-(t_5)}ud = \{d_{11}\}$,
 $d_{13-(1)}ud = \{d_4\}$, $d_{14-(1)}ud = \{d_5\}$, $d_{15-(1)}ud = \emptyset$, $d_{15-(t_6)}ud = \{d_{14}\}$, $d_{16-(1)}ud = \{d_5\}$,
 $d_{17-(1)}ud = \{d_4\}$, $d_{18-(1)}ud = \emptyset$, $d_{18-(t_8)}ud = \{d_{17}\}$, $d_{19-(t_7)}ud = \{d_{16}\}$, $d_{19-(1)}ud = \emptyset$.
 $d_{19-(t_9)}ud = \{d_{18}\}$, $d_{20-(1)}ud = \{d_4\}$

1) (1) $i = m - 1$

(5) $s_1 = 4 * i$

(9) $t_1 = s_1$

(13) $s_2 = s_1 - 4$

(17) if ($i > j$) goto 7

(21) $t_8 = s_2$

(25) $a[t_0] = x$

(29) $t_{12} = 4 * i$

(33) $t_{15} = 4 * n$

(2) $j = n$

(6) $s_2 = 4 * j$

(10) $t_3 = a[t_2]$

(14) $t_4 = s_2$

(18) $t_6 = s_1$

(22) $t_9 = a[t_8]$

(26) goto 7

(30) $t_{13} = 4 * n$

(34) $a[t_{15}] = x$

(3) $t_1 = 4 * n$

(7) $i = i + 1$

(11) if ($t_3 < v$) goto 7

(15) $t_5 = a[t_4]$

(19) $x = a[t_6]$

(23) $a[t_7] = t_9$

(27) $t_{11} = 4 * i$

(31) $t_{14} = a[t_{13}]$

(35) $a[t_{15}] = x$

(4) $v = a[t_1]$

(8) $s_1 = s_1 + 4$

(12) $j = j - 1$

(16) if ($t_5 > v$) goto 12

(20) $t_7 = s_1$

(24) $t_{10} = s_2$

(28) $x = a[t_{11}]$

(32) $a[t_{12}] = t_{10}$

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第 _____ 页

J) 对循环优化后的基本块局部优化, 得

$$(27) \quad t_{11} = 4 * 1$$

$$t_{12} = t_{11}$$

$$t_{13} = 4 * n$$

$$t_{15} = t_{13}$$

$$a[t_{15}] = a[t_{11}]$$

$$a[t_{12}] = a[t_{13}]$$

联系方式: _____